

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VII

Roll No. :S030	Name:Pradeep Rajbhar
Class :SYIT	Batch : 01
Date of Assignment :05/08/2026	Date/Time of Submission :05/08/2026

Hash Table:

Write a program to:

- a. Implement a hash table with separate chaining for collision handling.

Code:

```
prac7.py - C:/Users/itlab/Desktop/S030/PYTHON ON DS/prac7.py (3.14.6)
File Edit Format Run Options Window Help

size =5
table = [[] for _ in range(size)]

def insert(key,value):
    table[(key%size)].append((key,value))

insert(10,"divyanshu")
insert(15,"Sahil")
insert(18,"Abhishek")
insert(25,"Dhruv")
insert(31,"Sumit")

print(table)
```

Output:

```
===== RESTART: C:/Users/itlab/Desktop/S030/PYTHON ON DS/prac7.py
>>> [[(10, 'divyanshu'), (15, 'Sahil'), (25, 'Dhruv')], [(31, 'Sumit')], [], [(18, 'Abhishek')], []]
```

b. Store and retrieve data from the hash table.

Code:

```
*prac7.py - C:/Users/itlab/Desktop/S030/PYTHON ON DS/prac7.py (3.14.6)
File Edit Format Run Options Window Help

size =5
table = [[] for _ in range(size)]

def insert(key,value):
    table[(key%size)].append((key,value))

def search(key):
    for k,v in table [key%size]:
        if k==key:
            return v
    return "Not Found"

insert(10,"Divyanshu")
insert(15,"Sahil")
insert(18,"Abhishek")
insert(25,"Dhruv")
insert(31,"Sumit")

print(table)

key = int(input("Enter Key:"))
print(search(key))

print("Pradeep S030")
```

Output:

```
===== RESTART: C:/Users/itlab/Desktop/S030/PYTHON ON DS/prac7.py
[[ (10, 'Divyanshu'), (15, 'Sahil'), (25, 'Dhruv') ], [ (31, 'Sumit') ], [], [ (18, 'Abhishek') ], [] ]
Enter Key18
Abhishek
Pradeep S030
\\
```

